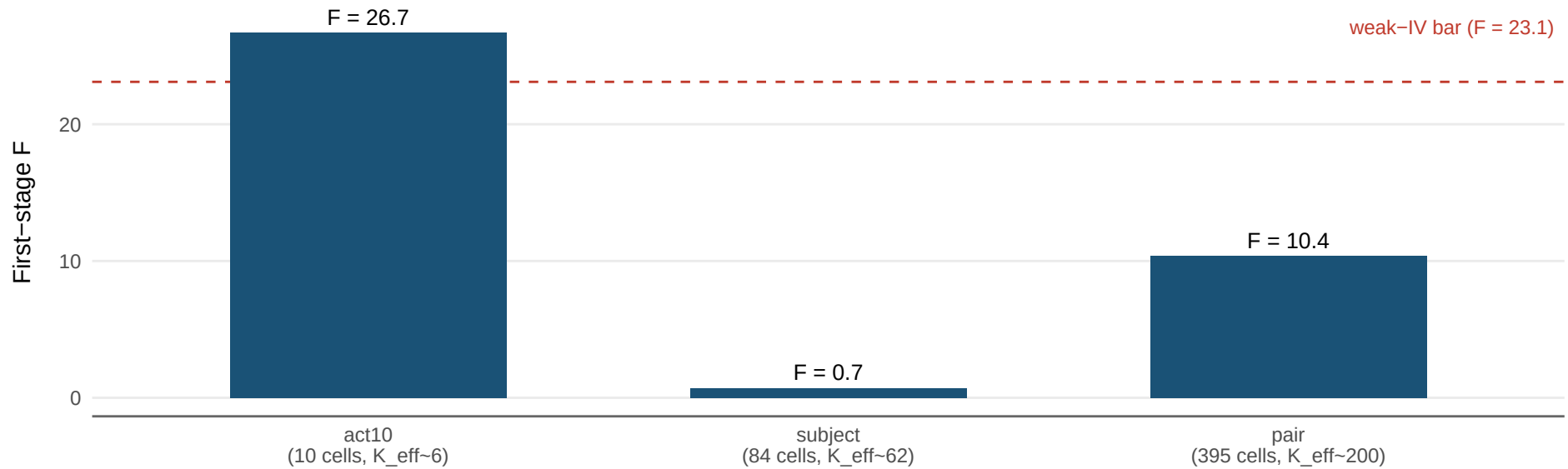
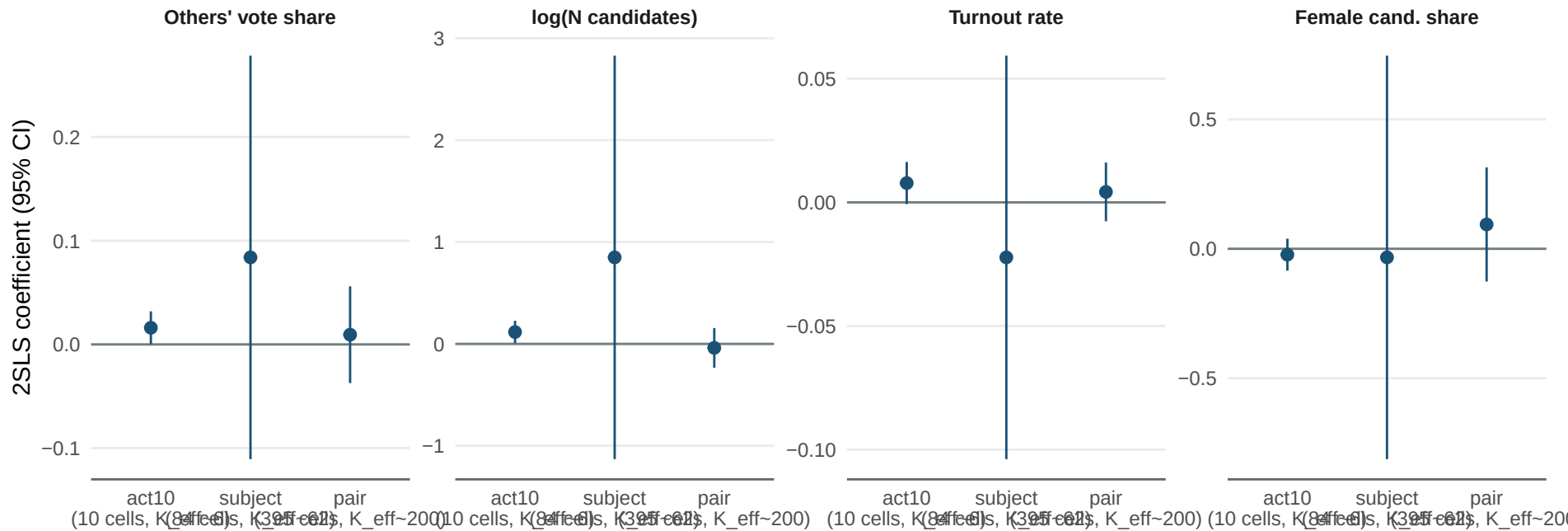


A. First-stage relevance collapses as the shock partition gets finer



B. 2SLS estimate stays near zero, but its CI explodes once relevance is gone



Each finer partition raises the effective number of shocks ($K_{\text{eff}} 6 \rightarrow 62 \rightarrow 200$) but destroys first-stage relevance ($F 26.7 \rightarrow 0.7 \rightarrow 10.4$), because sparse class $\times$ subject cells yield noisy national shocks. The null is grain-invariant; relevance is not. Mirror image of Ash-Morelli-Vannoni (JPE 2025), whose dense legislative topics keep relevance at every K in $\{6, \dots, 48\}$.